Set 1

- **Question A:**

  Scenario: Design a static 3-page HTML website for a *Central University Library*. The site must include a Homepage with semantic layout tags (`<header>`, `<nav>`, `<main>`, `<footer>`), an Online Catalog page displaying a structured table of available books (ISBN, Title, Author, Status), and an About/Contact page with nested lists of library rules and operating hours.

- **Question B:**

  Scenario: Develop an interactive *E-Commerce Shopping Cart* using HTML and JavaScript DOM events. Create dynamic buttons to add products to the cart, update item quantities, remove items from the DOM dynamically using event listeners, and compute the grand total in real time.

Set 2

- **Question A:**

  Scenario: Build an HTML5 *Tech Conference Event Registration Form*. The form must incorporate appropriate HTML5 input types (`email`, `tel`, `url`, `date`, `number`), range sliders for experience level, `<datalist>` for selecting tech tracks, `<fieldset>` grouping, and native validation attributes (`required`, `pattern`, `min`, `max`).

- **Question B:**

  Scenario: Create a responsive *Online Learning Portal* authentication layout. Build a top navigation bar with a logo, menu links, and call-to-action buttons using CSS Flexbox. Implement tabbed Sign-In and Sign-Up form cards below the navigation bar featuring styled inputs, hover effects, and custom CSS buttons.

Set 3

- **Question A:**

  Scenario: Design a responsive landing page for a *SaaS Cloud Analytics Product*. Use HTML5 and CSS Flexbox/Grid along with CSS Media Queries to adapt the layout across desktop (3 columns), tablet (2 columns), and mobile (1 column) viewports.

- **Question B:**

  Scenario: Implement an *Interactive Product Image Carousel* for an online shoe store using HTML, CSS, and JavaScript. The gallery must include "Next" and "Previous" controls, auto-play slide intervals, smooth CSS transition effects on image changes, and dynamic dot indicators that update active styles via JS.

Set 4

- **Question A:**

  Scenario: Create an HTML website for a *Fine Dining Restaurant*. The web page must display a top banner, an interactive navigation menu using anchor tags (`<a href="#section">`), an image gallery of signature dishes, an HTML table showcasing the multi-course menu with pricing, and embedded `<audio>` / `<video>` promotional clips.

- **Question B:**

  Scenario: Build a *Student Grade Performance Evaluator* web application. Construct an HTML interface where users enter student names and marks for 5 subjects. Write JavaScript logic embedded within HTML to calculate total marks, percentage, letter grade, and dynamically append result rows into an HTML table.

Set 5

- **Question A:**

  Scenario: Construct an HTML5 *Hospital Patient Admission Form*. Utilize specialized form controls including `<select>` dropdowns for department assignment, `<textarea>` for medical history, HTML5 `date` and `time` pickers, radio buttons for insurance status, and checkboxes for existing symptoms with built-in form validation.

- **Question B:**

  Scenario: Build a *Dynamic Task Management Application* using JavaScript DOM elements. Users should be able to type a task into an input field and click "Add". Implement event listeners to mark tasks complete (strikethrough styling), filter completed vs. pending tasks, and delete individual task items from the list.

Set 6

- **Question A:**

  Scenario: Create a responsive portfolio website for a *Freelance Web Developer*. Implement a responsive CSS Grid project gallery that resizes fluidly without breaking, sticky headers, styled call-to-action buttons, and mobile-friendly typography scaled with relative CSS units (`em`/`rem`/`vw`).

- **Question B:**

  Scenario: Develop a *Fitness Gym Membership Portal* authentication interface. Build a styled horizontal CSS navigation bar with drop-down menus. Below it, design split Sign-In and Sign-Up forms using CSS Grid, styled focus borders, gradient buttons, and responsive breakpoints.

Set 7

- **Question A:**

  Scenario: Build a static HTML web page for a *City General Hospital*. Structure the document semantically with sections for emergency alerts, specialized departments listed via unordered bullet points, operating hours displayed in an HTML table, and iframe-embedded Google Maps for directions.

- **Question B:**

  Scenario: Build a dynamic *Light/Dark Theme Switcher* for a blog site using HTML, CSS, and JavaScript. Create a toggle button that modifies CSS custom variables dynamically, smoothly animates background color transitions, transforms button icon positions, and updates font colors across the entire web page.

Set 8

- **Question A:**

  Scenario: Design an HTML5 *Flight Ticket Booking Form* for an airline portal. Include fields for trip type (one-way/round-trip radio options), departure/destination airports using `<datalist>`, passenger counts via `<input type="number">`, departure date limits using `min` and `max` constraints, and seat selection dropdowns.

- **Question B:**

  Scenario: Create an interactive *Real-Time Currency Converter* web application. Build an HTML layout with target currency selection dropdowns and amount input fields. Use JavaScript embedded in HTML to execute rate conversions dynamically on keyup/change events and update the results on screen without reloading.

Set 9

- **Question A:**

  Scenario: Design a responsive *Subscription Pricing Page* for a streaming media platform. Create three tiered pricing cards (Basic, Standard, Premium) using CSS Flexbox. Highlight the recommended plan using CSS scaling, and use media queries to stack the cards vertically on smaller mobile screens.

- **Question B:**

  Scenario: Build an *Interactive Online Quiz Application* using JavaScript DOM events. Display questions dynamically one at a time, capture user selection using click event listeners, track real-time score counters, highlight correct/incorrect answers instantly, and display final results upon submission.

Set 10

- **Question A:**

Scenario: Construct an HTML product showcase page for an *Electronics Brand*. Include structured headings, product description paragraphs, image lists with `alt` attributes, tabular specification sheets comparing device models, and anchor link navigation to jump between sections.

- **Question B:**

Scenario: Design a *FinTech E-Banking Portal* login screen. Build a CSS navigation bar containing brand logo, security links, and help desk contact. Create side-by-side Customer Sign-In and New User Registration cards styled with CSS box-shadows, rounded borders, input icon alignments, and hover states.

Set 11

- **Question A:**

Scenario: Develop an HTML5 *Job Portal Application Form*. Requirements include `<input type="file">` restricted to PDF uploads (`accept=".pdf"`), an HTML5 experience rating slider (`<input type="range">`), custom input regex pattern validation for phone numbers, and interactive text areas with max character limits.

- **Question B:**

Scenario: Create a JavaScript *Dynamic Modal Popup System* for an online store checkout alert. Trigger a hidden modal overlay with fade-in and scale animations when a user clicks "Proceed to Payment". Include a close button (`×`) and background backdrop click listener that smoothly animates the modal closed.

Set 12

- **Question A:**

Scenario: Design a responsive news platform layout featuring a multi-column main news section, sticky sidebar, and responsive gallery grid. Apply CSS Media Queries to transition the site seamlessly between wide screens, tablets, and mobile screens.

- **Question B:**

Scenario: Develop an *Event Countdown Timer & Digital Clock* using HTML and JavaScript. Display live hours, minutes, and seconds, alongside a timer counting down to a specific launch date. Update the DOM displays every second using JavaScript `setInterval()` functions.

Set 13

- **Question A:**

Scenario: Design a static HTML web page for a *Music & Cultural Festival*. The page must include embedded audio previews of performing artists, promotional video trailers

(`<video>`), a daily schedule layout rendered via HTML tables, and external hyperlinks to ticket sponsors.

- **Question B:**

  Scenario: Implement a *Real-Time Table Search and Filter System* using JavaScript DOM events. Build an HTML table of employees (ID, Name, Department, Designation). Write JS functions using `keyup` event listeners to match input query text and instantly hide non-matching HTML table rows.

Set 14

- **Question A:**

  Scenario: Build an HTML5 *Customer Feedback & Survey Form* for a hotel chain. Utilize specific HTML5 elements including `<input type="color">` for theme rating, `<input type="url">` for social media profiles, datalists for city selection, rating radio buttons, and clear/submit buttons.

- **Question B:**

  Scenario: Develop a *Real Estate Listing Portal* access page. Construct a responsive CSS top navigation bar with drop-down menus for property categories. Below it, create centered CSS Sign-In and Sign-Up forms with styled floating input labels, password visibility icons, and hover animations.

Set 15

- **Question A:**

  Scenario: Design a responsive *Online Magazine Article Page*. Use CSS Grid for the layout structure (header, main article content, sidebar, footer). Ensure responsive media scaling for all inline photos and video containers across all display resolution sizes.

- **Question B:**

  Scenario: Build an *Accordion FAQ Section* for a software help desk using HTML, CSS, and JavaScript. Clicking a question panel heading should expand/collapse the answer container with smooth CSS height/opacity transition animations while closing other open accordion items dynamically.

Set 16

- **Question A:**

  Scenario: Construct a multi-page HTML site for a *Travel & Tourism Agency*. Include a homepage introducing popular tours, a destination page with image thumbnails and descriptions, a tour itinerary table with pricing, and a contact page with HTML lists and contact details.

- **Question B:**

  Scenario: Build an interactive *Restaurant Bill & Tip Split Calculator* using HTML and JavaScript. Provide input fields for Total Bill Amount, Tip Percentage, and Number of People. Dynamically calculate and display total tip, total bill with tip, and per-person breakdown upon button click or input modification.

Set 17

- **Question A:**

  Scenario: Create an HTML5 *Auto Insurance Quote Generator Form*. Incorporate form controls with regex validation for vehicle registration formats, `<fieldset>` sections for personal vs. vehicle data, `<datalist>` auto-suggestions for vehicle models, date pickers for policy start date, and policy selection radio cards.

- **Question B:**

  Scenario: Create a *Live Social Media Post Composer* using JavaScript DOM elements. Include an HTML `<textarea>` where users type a post. Dynamically calculate remaining characters (out of 280), highlight the counter in red when exceeding the limit, and disable the "Publish" button using DOM property manipulation.

Set 18

- **Question A:**

  Scenario: Design a responsive *Photography Gallery Page*. Implement a CSS Grid layout with image hover animations (zoom/overlay text effect). Use CSS Media Queries to switch layout columns from 4 on desktop down to 2 on tablet and 1 on mobile viewports.

- **Question B:**

  Scenario: Build a *SaaS Application Dashboard Login Layout*. Construct a top CSS navigation bar equipped with a responsive hamburger toggle menu. Integrate split Sign-In and Sign-Up forms featuring CSS smooth tab switching, shadow elevation styles, and validation highlight borders.

Set 19

- **Question A:**

  Scenario: Create a static HTML web page for a *College Computer Science Department*. Build a structured layout containing faculty directory tables, syllabus bullet lists, upcoming event announcements, and embedded HTML maps for campus location.

- **Question B:**

Scenario: Develop a *Dynamic Scroll-Header & Back-to-Top Button* feature using JavaScript styles and animations. As the user scrolls down the page, change the navigation bar's background color, height, and box-shadow via JS, while smoothly fading in a floating "Scroll to Top" button that scrolls smoothly back to the page top when clicked.

Set 20

- **Question A:**

  Scenario: Create an HTML5 *University Course Registration Form*. Integrate input validation for student ID numbers (`pattern`), dropdown lists for department selection, HTML5 `<datalist>` for course elective search, date picker for semester start, and agreement checkbox constraints.

- **Question B:**

  Scenario: Build an interactive *Body Mass Index (BMI) Health Calculator* using HTML and JavaScript. Capture height (cm) and weight (kg) inputs. Calculate the BMI value upon button click, update the HTML display with the calculated score, and dynamically change the CSS background color of the result container based on health category (Underweight, Normal, Overweight, Obese).